

Science 2 -Assignment (2026)

(Total Marks 32)

Submit on or before 14th February, 2026

Q1. Perform integration using the Monte Carlo method* for the functions applicable to you. The functions you need to use is your Roll Number % 5 + 1. Therefore, if your Roll Number is 2021101110, take option 1. Similarly for 2021101113, take option 4. Attach your script as well.

1. (a) $\sqrt{\tan x}$ in the range $[0, \pi/4]$ (b) $x \sin x$ in the range $[-\pi, \pi]$
2. (a) $\sqrt{\cot x}$ in the range $[\frac{\pi}{4}, \frac{\pi}{2}]$ (b) $x^2 \sin x$ in the range $[-\pi/2, \pi/2]$
3. (a) e^x in the range $[-3, 3]$ (b) $e^x \sin x$ in the range $[-1, 1]$
4. (a) $x^2 + 4x$ in the range $[-1, 1]$ (b) $x^3 \cos x$ in the range $[-\pi/2, \pi/2]$
5. (a) $1 + ax$ in the range $[-5, 5]$ (b) $\cos 3x - \cos 2x$ in the range $[0, \pi]$

(*Plot the integration value as a function of N (number of samples) where $N = 10, 10^2, 10^3, 10^4$).

(5+5 marks)

Q2. Take a graph with N nodes and N+3 links such that graph is connected. Calculate the leading eigenvector using power method. The associated graph is symmetric and binary, and N = is your Roll Number % 5 + 5. Submit the code. **(5 Marks)**

Q3. You need to simulate 1D Random Walks with N timesteps with step size of 1 in each timestep. The person has equal probabilities of moving in either direction, i.e, at each step the person can move -1/+1.

i) A drunk person starts out at initial position “a”. Plot the probability that the person returns back to the origin after N timesteps for different values of N in the range of 1 to 100. **(3 marks)**

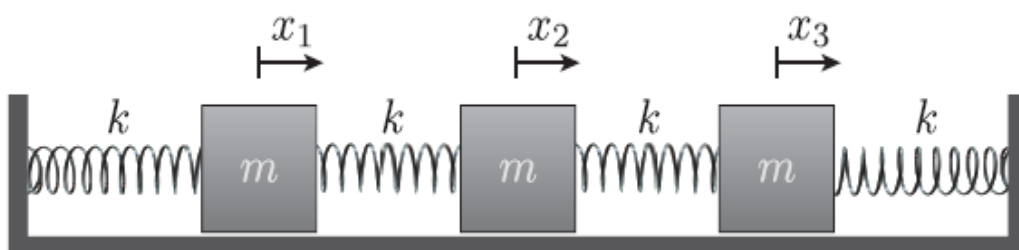
ii) Now assume, two drunk people start out at initial positions “a” and “b” respectively. Plot the probability that they meet again after N timesteps for different values of N in the range of 100 to 1000, with the increment of 50. **(4 marks)**

To compute probabilities simulate the experiments appropriate number of times (at least 1000).

$a = -(\text{Roll Number \% } 5 + 1), \quad b = 2 * (\text{Roll Number \% } 10)$

Therefore, if your Roll Number is 2021101132, then $a = -3, b = 4$.

Q4.



- i) Find three normal modes and the associated eigenvectors.
- ii) Write a script (based on Euler method) to explain the motion of each block corresponding to each mode. (Follow my PPT to choose the initial conditions wisely. Example:: **set 1:** $x_1(t=0)=0.5$, $x_2(t=0)=0$; and $x_3(t=0)=-0.5$; **set 2:** $x_1(t=0)=0.5$, $x_2(t=0)=0$; and $x_3(t=0)=0.5$, and so on. Set the initial velocity of each block to zero.)
- iii) Try to find the frequencies from the observed signals, and check whether they are connected to the original signals.

Take $k = (\text{Roll Number \% } 4) + 1, m = 1$

(4+4+2=10 marks)